JJJ-V-002173A(Army-GL)
July 8, 1985
SUPERSEDING
JJJ-V-002173(Army-GL)
22 June 1983

INTERIM FEDERAL SPECIFICATION

CARROTS, SLICED, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers sliced carrots, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495	- Coatings, Exterior, for Tinned Food Cans
PPP-B-636	- Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

FSC 8915

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specifications for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR parts 1-199)

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture (USDA)

U.S. Standards for Grades of Canned Carrots
U.S. Standards for Grades of Frozen Carrots

Regulations Governing Inspection and Certification of Processed Fruits and Vegetables, Processed Products Thereof, and Certain Other Processed Food Products

(Application for copies should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Association of Official Analytical Chemists (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1 and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Carrots, fresh. The fresh carrots shall be a variety suitable for canning. The fresh carrots shall be well trimmed, firm, clean, of typical color, smooth and well formed. They shall be free from secondary new top growth, soft rot and free from damage caused by freezing, growth cracks, sunburn, pithiness, woodiness, internal discoloration, oil spray, dry rot, other diseases, insects or other means. The diameter of each carrot shall not be less than 3/4 inch or more than 1-1/2 inch and the length not less than 5 inches.

3.2.2 Carrots, sliced, frozen. The carrot slices shall be U.S. Grade A of the U.S. Standards for Grades of Frozen Carrots and shall be of the latest season's crop. The carrot slices shall be 0.25 to 0.50 inches in thickness and have a diameter in the range of 0.75 to 1.50 inches.

3.2.3 Salt. Salt shall be non-iodized, white, refined sodium chloride, with or without anticaking agents.

3.3 Carrot preparation.

3.3.1 Fresh carrot preparation. The carrots shall be washed thoroughly to remove soil and foreign material, and sorted to remove any which are diseased or otherwise damaged. The sound stock shall be peeled and trimmed to remove discolored, damaged or undesirable portions. They shall be sliced to a thickness of 0.25 to 0.50 inch and properly blanched if blanching had not been accomplished during the peeling operation. The blanched carrots shall be immediately cooled in cold water before canning. When firming is necessary,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.

it shall be in accordance with the procedure specified in the Standards of Identity for Canned Vegetables, except that only calcium chloride shall be used as the firming agent.

3.3.2 Frozen carrot preparation. Frozen carrots shall be thawed to the extent necessary for proper tray pack can filling.

3.4 Brine preparation. Twenty-two pounds of salt shall be placed in proper container and water added until 100 gallons of brine is made.

3.5 Tray pack filling and sealing. Each tray pack can shall be filled with product so as to conform to the following requirements:

a. Each tray pack can (see 5.1.1) shall be filled with carrots to meet the drained weight requirements of 3.7 and the can filled with brine to not less than 102 ounces.

b. Brine may be filled hot or cold. Brine shall be made fresh daily and brine remaining at the end of the day shall be discarded.

c. The cans shall be hermetically sealed with a minimum of 10 inches (Hg) vacuum immediately after filling.

3.6 Tray pack thermoprocessing. The filled and sealed cans shall be thermostabilized by retorting at 250°F or below until a sterilization value (Fo) of not less than 4.2 has been achieved.

3.7 Finished product requirements. The finished product shall comply with the following:

a. The product shall meet the requirements for U.S. Grade A, Sliced, Canned Carrots of the U.S. Standards for Grades of Canned Carrots except for size and texture. No slice shall be greater than 1-1/2 inches in diameter. Texture shall be U.S. Grade C or better of the U.S. Standards for Grades of Canned Carrots.

b. The average net weight shall be not less than 102 ounces.

c. No individual can shall contain less than 100 ounces of product.

d. No individual can shall contain less than 69 ounces drained weight of sliced carrots.

e. No individual can shall have a salt content of more than 1.5 percent nor less than 0.5 percent.

3.7.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.

3.8 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods, and the plant sanitation requirements of the appropriate Government inspection agency.

3.9 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act, and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or other applicable purchase document.

4.4.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.

actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 empty tray pack cans and lids and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 102 ounces or if the net weight of any individual can is less than 100 ounces, the lot shall be rejected.

4.6.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be opened and inspected for the defects listed in table II.

TABLE I. Double sampling plan for product inspection. 1/

Lot size (cans)	Sample size (No. of cans)	Cumulative sample	Acceptance number	Rejection number
0-3200	8	--	0	2
	8	16	1	2
3201-35000	13	--	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.

TABLE II. Product inspection 1/ 2/ 3/

Category	Defect
Major	
101	Drained weight of carrots is less than 69 ounces

1/ The presence of any foreign material (e.g., glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale, etc.) or foreign color shall be cause for rejection of the lot.

2/ Product not meeting grade type, size and/or style shall be cause for rejection of the lot.

3/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

4.6.3 Salt content testing. Nine tray packs shall be selected from the lot and distributed as follows:

- three for laboratory analysis
- three for submission to the contractor
- three for retention by the AMS agent as a reserve sample

The three tray packs for laboratory analysis shall be individually tested for salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, except that preparation shall be as follows. The cans shall be opened and the entire contents of each can shall be separately blended in a Warning blender or equivalent. The lot shall be rejected if the salt content in any individual tray pack is greater than 1.5 percent or less than 0.5 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the salt content requirement. When the reserve sample is analyzed, the analyses for salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: (1) National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible; (2) Reinforcement with nonmetallic strapping or tape is not used. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above top edge of box; (4) Length or width of pads less than box dimensions by more than 1/8 inch; (5) Arrangement or number of cans not as specified.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 102 ounces of food product, sealed, and thermoprocessed as specified in 3.5, 3.6 and 3.7. The base plate of the can shall be electrolytic chromium-plated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated over all on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that the cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be not less than 1/8 inch of the full length and width dimensions of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.

box and shall be fabricated of grade V3c fiberboard. The pads shall be placed between the cans and on the top and bottom of the stacked cans. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimension of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. In addition, the name of product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.7 after submerging the can in boiling water for 30 minutes. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 10 to 15 minutes. Avoid overheating to the point where internal pressure is evident (can shows evidence of bulging). Remove from water and open lid for serving. Caution: Use care when opening as pressure may have been generated within the can.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F.

WARNING: Do not place closed can in oven, as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 25 servings of 1/2 cup each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary markings in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.3, and 6.3).
(c) Provisions for approved preproduction samples (see 3.7.1 and 6.3).
(d) Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).
(e) Level of packaging and packing required (see 5.1 and 5.2).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection, selection, and approval of the first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.

Preparing activity:

Army-GL

Project No. 8915-A789

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

US ARMY NATICK RESEARCH
and DEVELOPMENT CENTER
ATTN STRNC-ES
Natick, MA 01760-5014

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research
and Development Center
ATTN STRNC-ES
Natick, MA 01760-5014

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
JJJ-V-002173A (Army-GL)	Carrots, Sliced, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation:

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

NATICK OP 20 Nov 84

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:45Z
Check the source to verify that this is the current version before use.